Introduction to Multimodal AI

Course Title: Multimodal AI

Course Code: AAI3031-01

Dr. Khan Muhammad (칸무하마드)

Associate Professor

Department of Applied Artificial Intelligence

Sungkyunkwan University

Email: khanmuhammad@g.skku.edu



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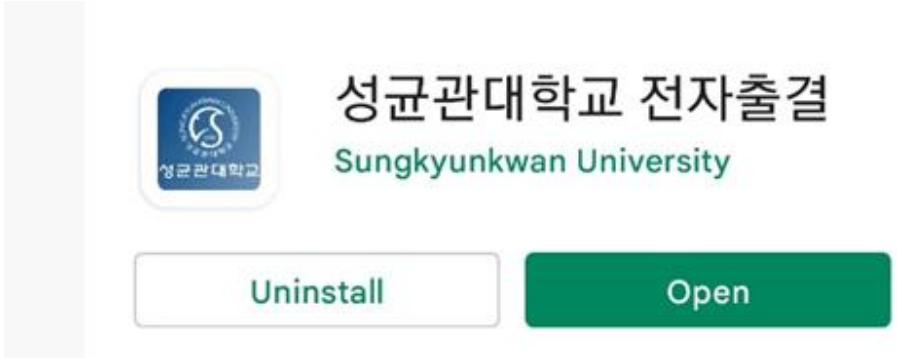
1. Course Information

- **3 Credits** [Flipped Learning (1 hour online/recorded, 2 hours offline)]

  Lectures: 3 hours weekly

  - Room: International Hall Multimedia Lecture Room (**9B217**)
    - Thursday 15:00 to 15:50 PM
    - Thursday 16:00 to 16:50 PM
    - Total: 1 Hour, 40 min.

- **Type:** Theory/practical/invited speeches/Recorded/online/in-person
- **Attendance**
  - "SKKU-Attend" for attendance, absence and late rules
  - Default setting for the time being
  - Zoom/Manual



2. Students' Assessment

- Attendance: 10%
- Mid-term exam: 20%
- Final-term exam: 40%
- Practice Assignment (PA)/Quizzes/Homework etc.: 15%
- Presentation: 15%

- **Project**
  - Multimodal AI and your major related project completion
  - Survey/Research articles

- **Subjective**
  - Short-answer essay
  - Extended-response essay
  - Problem solving

- **Cheating/Copy Paste Policy**
  - Automatic F for both



3. Professor's Bio

Khan Muhammad (Senior Member, IEEE)



- Associate Professor (Tenure-Track), SKKU, Korea
- khanmuhammad@skku.edu
- +82-10-4831-2104
- Editorial Board Member, 7 Journals
- Area/Associate Editor, 7 Journals
- Top 2% Researcher (Stanford Uni. List), 2020-present
- #2 in Korea (Best CS Scientists 2026, Research.com)
- HCR (top 1%), 2021-2024

Publication Record

Over 300 articles/papers

As lead-author (1st or corresponding author)	**180+**
As co-author	**90+**
Total citations (Google Scholar)	**32230+ (H-Index: 93)**
Impact factor	**1750+**
Patents Record (Korean, US mainly)	**15**
Participation in Research Projects	**15**
Professional and Research Experience	**10+**

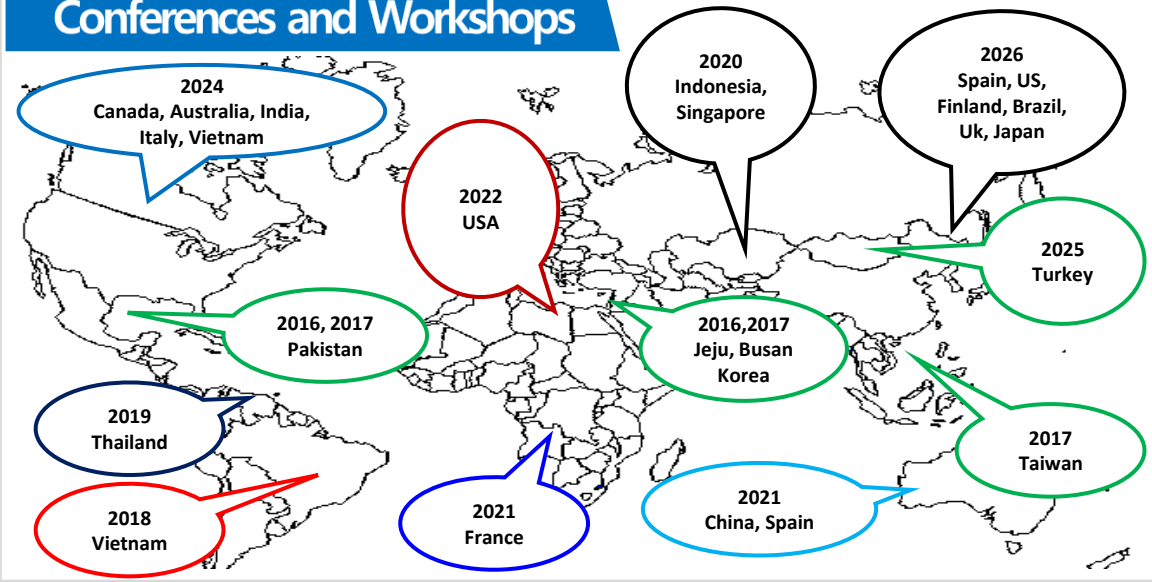
Research Interests

① **Computer Vision**
- Activity Recognition (Normal, Child Hazards, Aggression, Bullying)
- Image Enhancement (Low-light, Dehazing, Denoising)
- Person Re-identification

② **Large Language Models (LLMs)**
- Sentiment Analysis
- Hallucinations

③ **Intelligent Surveillance**
- Fire/smoke detection, segmentation and scene analysis
- Unexpected Road Hazards Detection and Analysis

Conferences and Workshops



4. Research Topics



5. Artificial Intelligence (AI)

5 (i): Intelligence Starts with the World

A reasonable question to think on it.

How Do We Make Sense of the World?

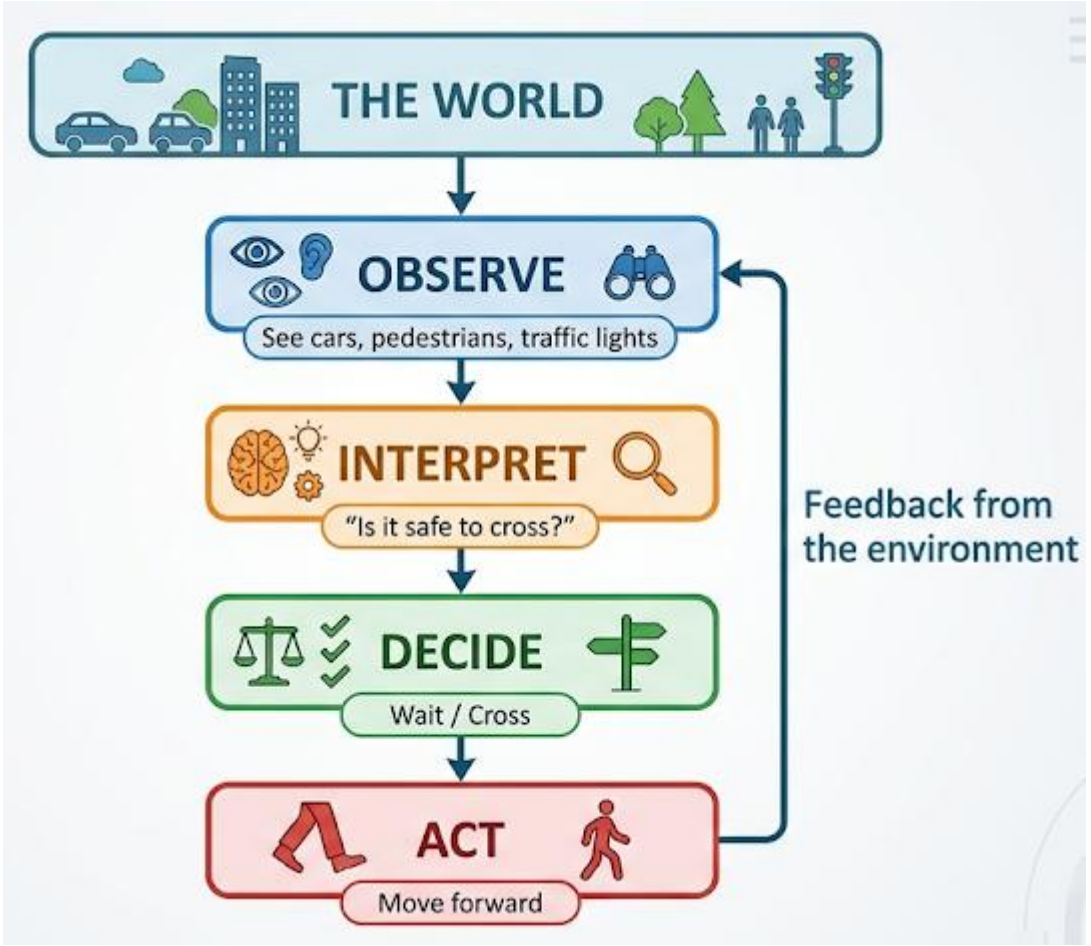
Observe → Interpret → Decide → Act

Example: A Human Crossing a Road

1. **Observe**
   See cars, pedestrians, traffic lights
2. **Interpret**
   "Is it safe to cross?"
3. **Decide**
   Wait / Cross
4. **Act**
   Move forward

A person can:

- Recognize a familiar face
- Understand spoken language
- Detect danger
- Learn from experience
- Choose an action

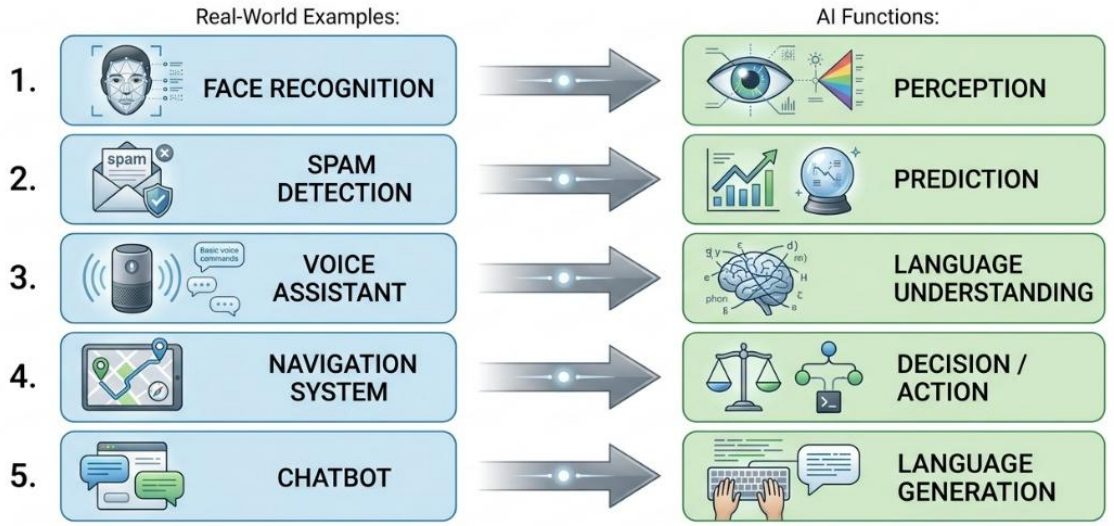


➢ **Intelligence begins with using information to understand a situation and make useful decisions.**

5 (ii): What Is Artificial Intelligence?

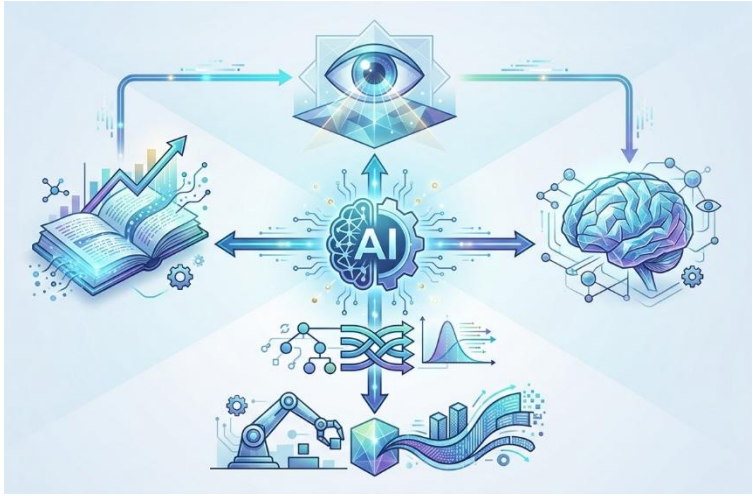
Definition:

Artificial Intelligence (AI) is the field of building computational systems that can perform tasks that normally require aspects of human intelligence.



AI can help machines:

- **Perceive**
  See and hear the environment
- **Learn**
  Learn patterns from data
- **Understand**
  Interpret language and information
- **Reason / predict**
  Use information to reach conclusions or predictions
- **Act / generate**
  Choose actions or produce new content



AI performs specific tasks that involve capabilities associated with intelligence. The example for it is given in the diagram.

5 (iii): Intelligence in Machines: Rules vs Learning

How Do We Teach a Machine to Solve a Problem?

We can explicitly program the solution OR we can give the machine examples and let it learn the pattern.

1:Explicit Rules	"Tell me what to do?"
2:Machine Learning	"Show me examples, So I can learn pattern"

Example:1

IF message contains suspicious pattern

↓

SPAM

ELSE

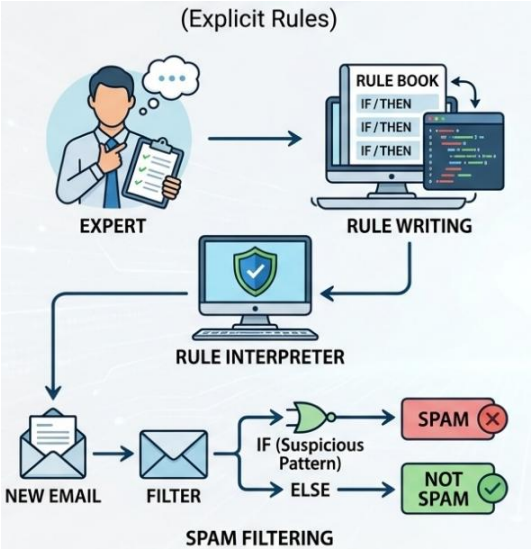
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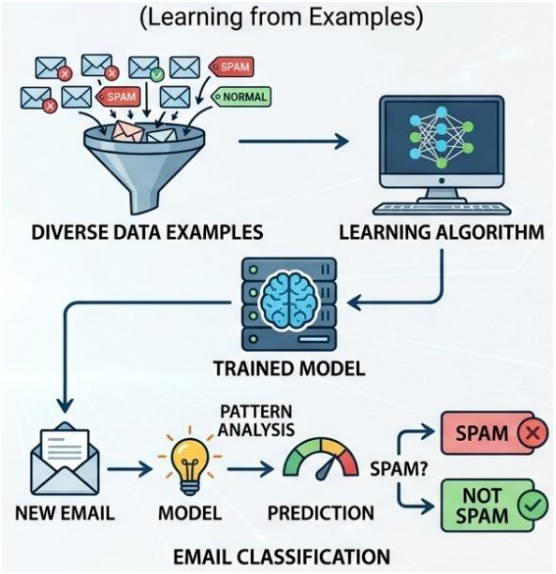
NOT SPAM

Example: 2

Many spam + normal emails

- Learning
- Learned model
- New email → Spam / Not Spam



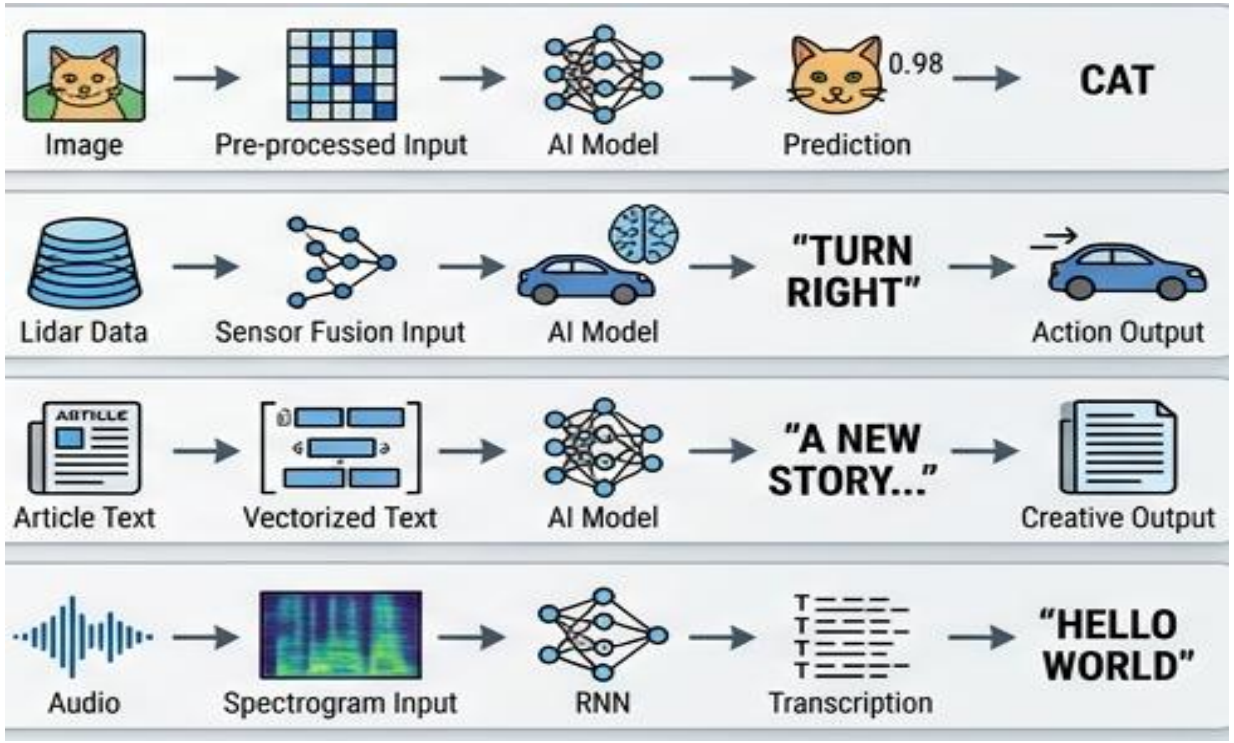


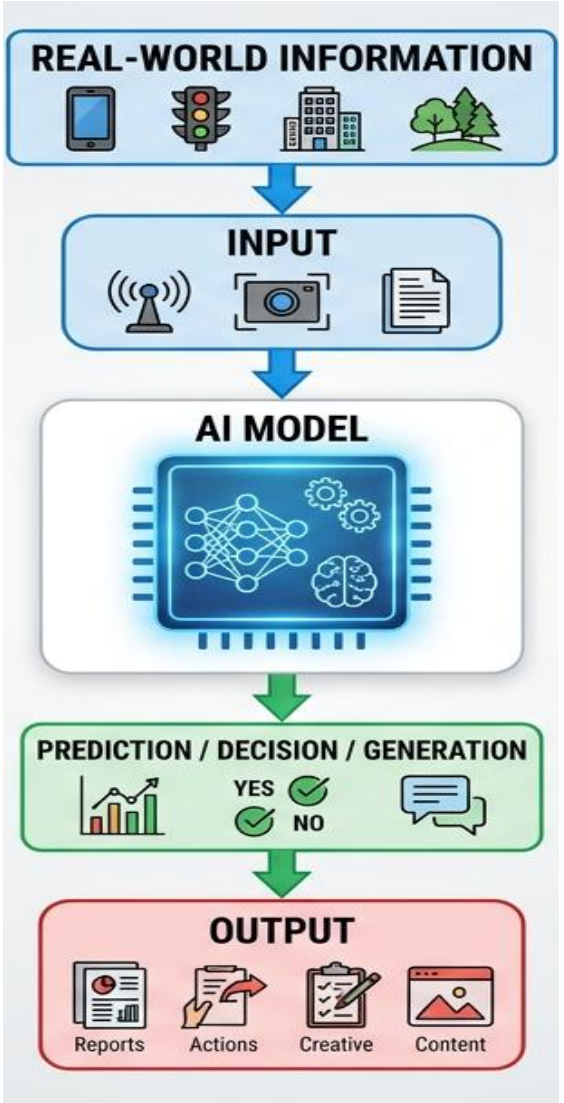
5 (iv): The Basic Idea of an AI System

Basic Idea



Some Visual Examples





- An AI system transforms input information into a useful output.

6. Multimodal AI

6 (i): The Real-World Problem

One event. Many signals.

- Imagine an autonomous vehicle approaching a busy crossing at night.

- A single real-world situation generates multiple, complementary (different but useful together) sources of information.

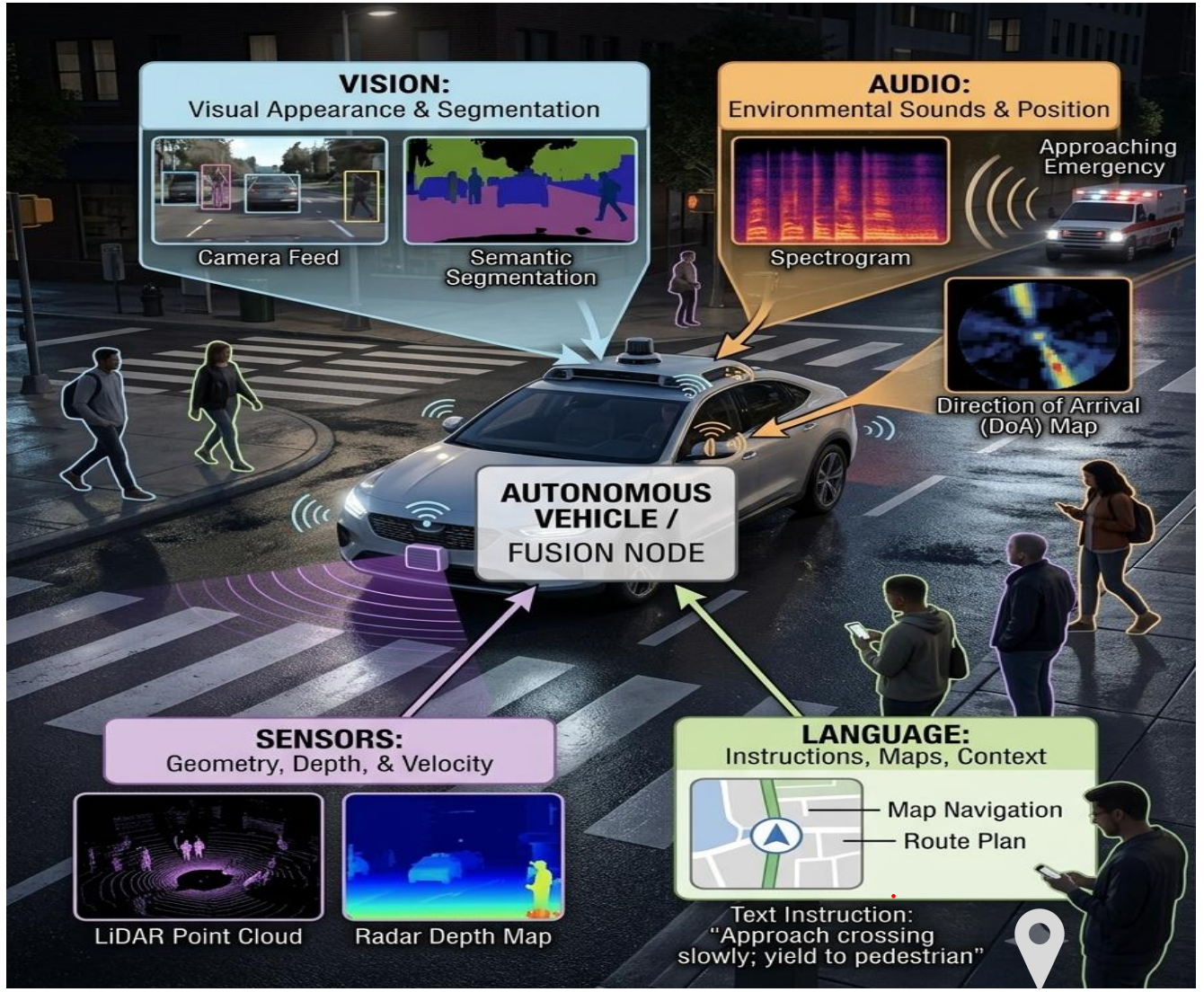
The challenge

Can one information source provide enough evidence to understand the situation reliably?

Real-world intelligence requires combining information that is distributed across different signals.

Think–pair–share

Which signal would you trust most in this situation?
When could that signal fail?



6 (ii): Why Multimodal AI?

The Central Question

How does intelligence emerge when information comes from multiple sources?

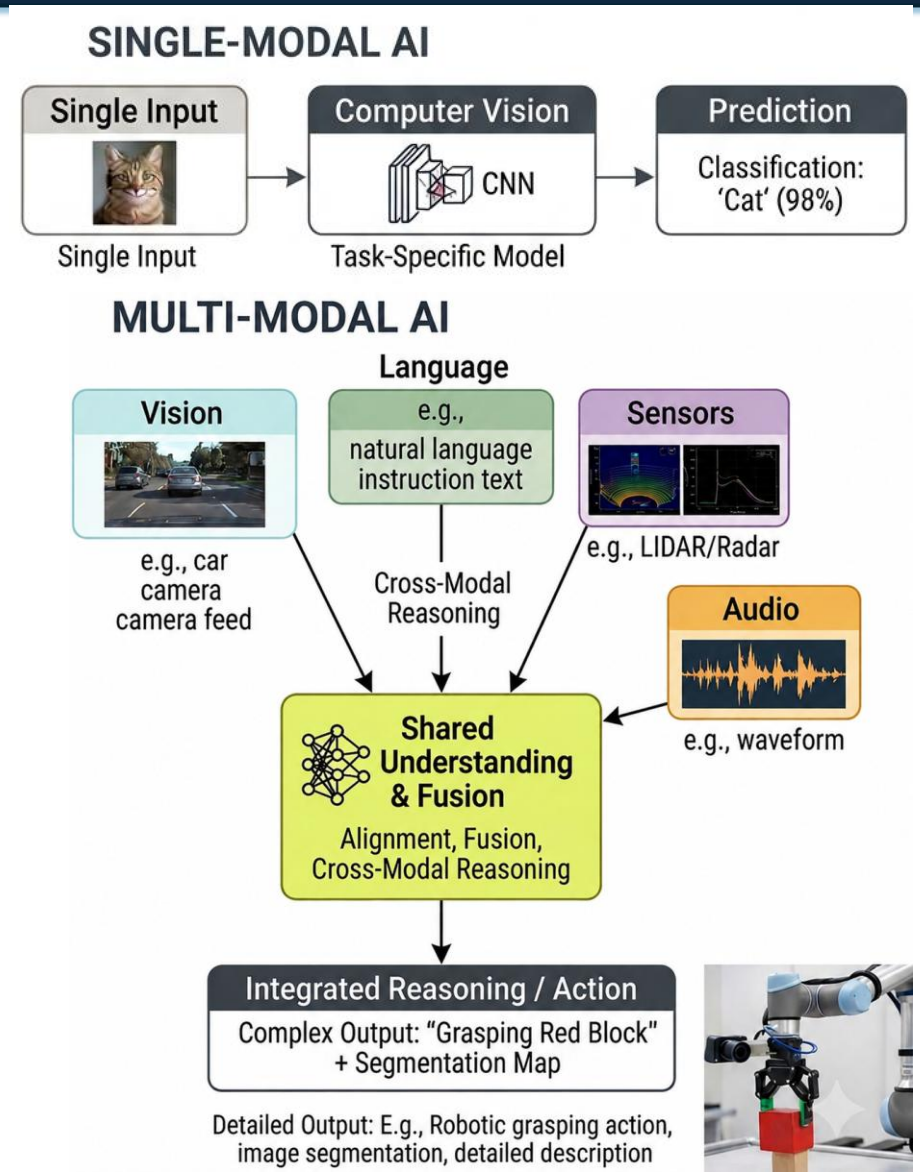
Traditional AI systems often specialize in one information type:

- **Computer Vision**
- **Natural Language Processing**
- **Speech AI**
- **Video Understanding**
- **Sensor AI**

But real-world intelligence rarely operates on one source alone.

Key idea

The real world is multimodal. Intelligent systems therefore need mechanisms for integrating heterogeneous information.

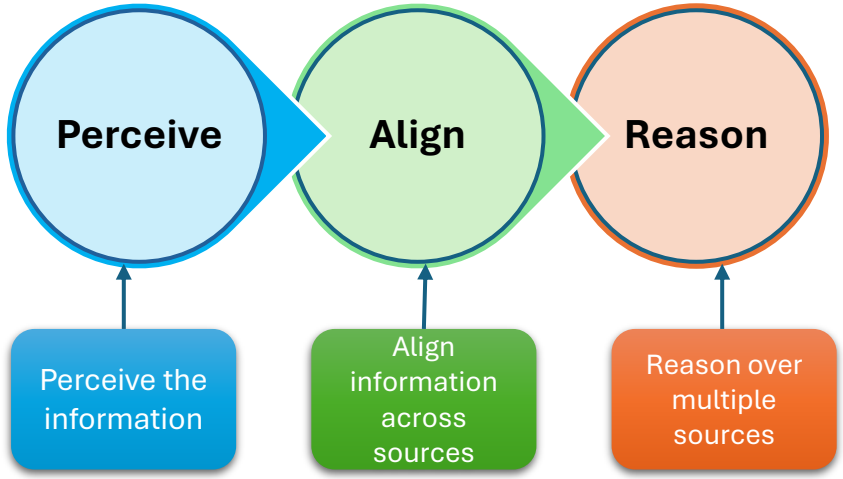


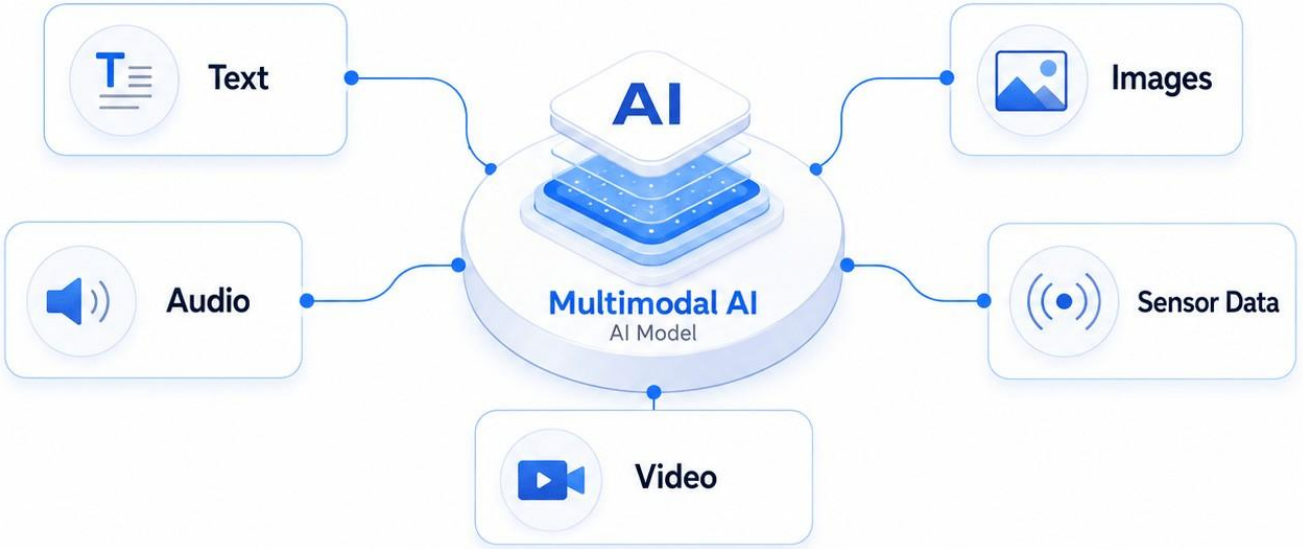
6 (iii): What is Multimodal AI?

Definition:

Multimodal AI refers to AI systems that can process, represent, connect, and reason over information originating from multiple modalities.

The course plan describes multimodal AI as systems that:





Important distinction

Multimodal AI is not simply putting several AI models together.

The central challenge is:

How can heterogeneous information be represented in a form that allows meaningful interaction between modalities?

6 (iv): Understanding "Modality"

Modality: A distinct channel or form through which information is represented or observed.

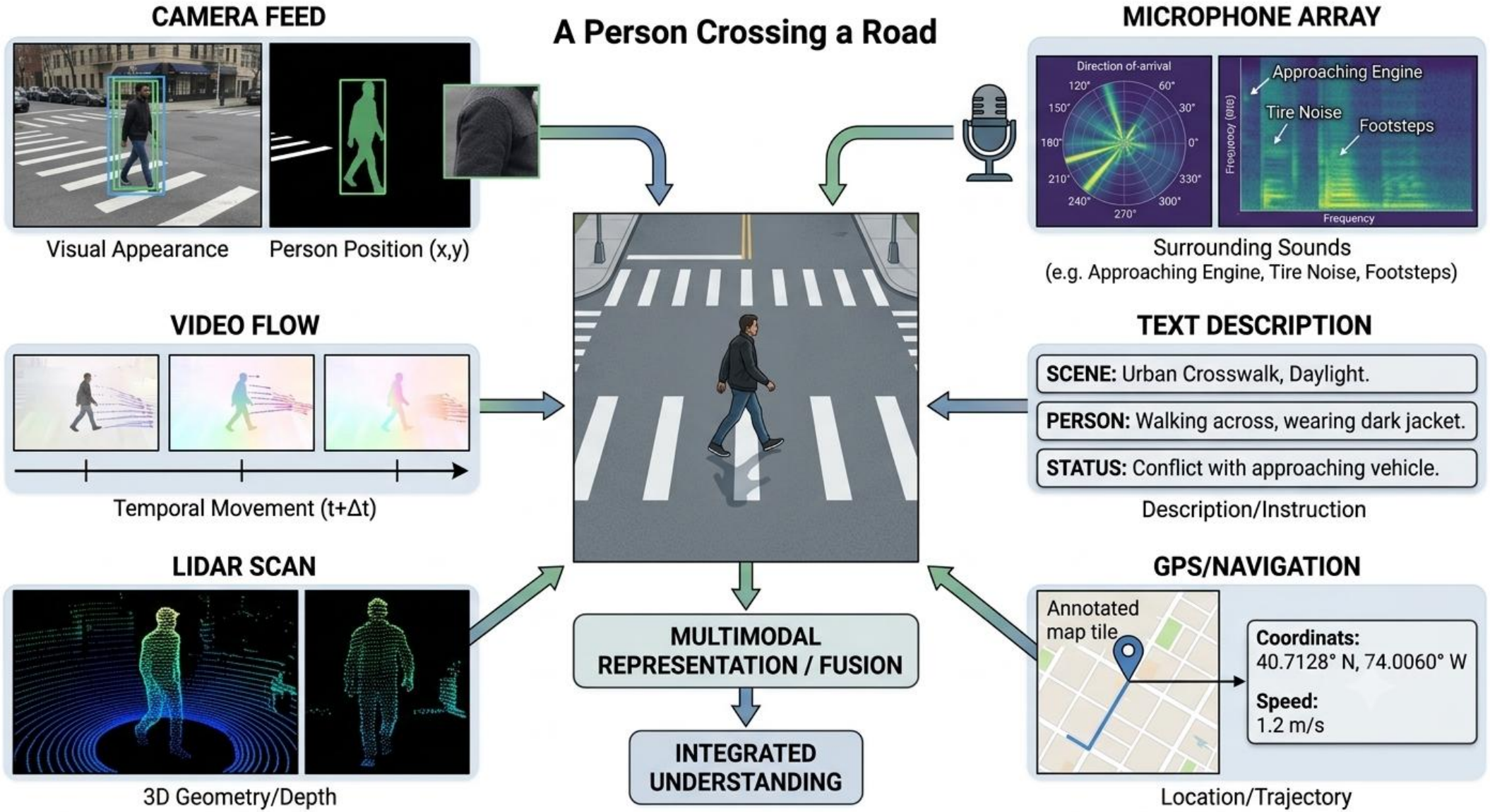
Common AI modalities:

Modality	Typical Data	Example
Vision	Images, pixels	Photograph
Text	Words, documents	Question
Audio	Waveforms, spectrograms	Speech
Video	Frames + time	Human Activity
Sensors	Measurements/signals	Temperature, IMU
3D / Spatial	Point clouds, depth	LiDAR
Structured Data	Tables/events	Patient Records

Key insight
Different modalities describe the same world from different information perspectives.

For example:
A person crossing a road
The *example is explained on the next slide* with visualization.

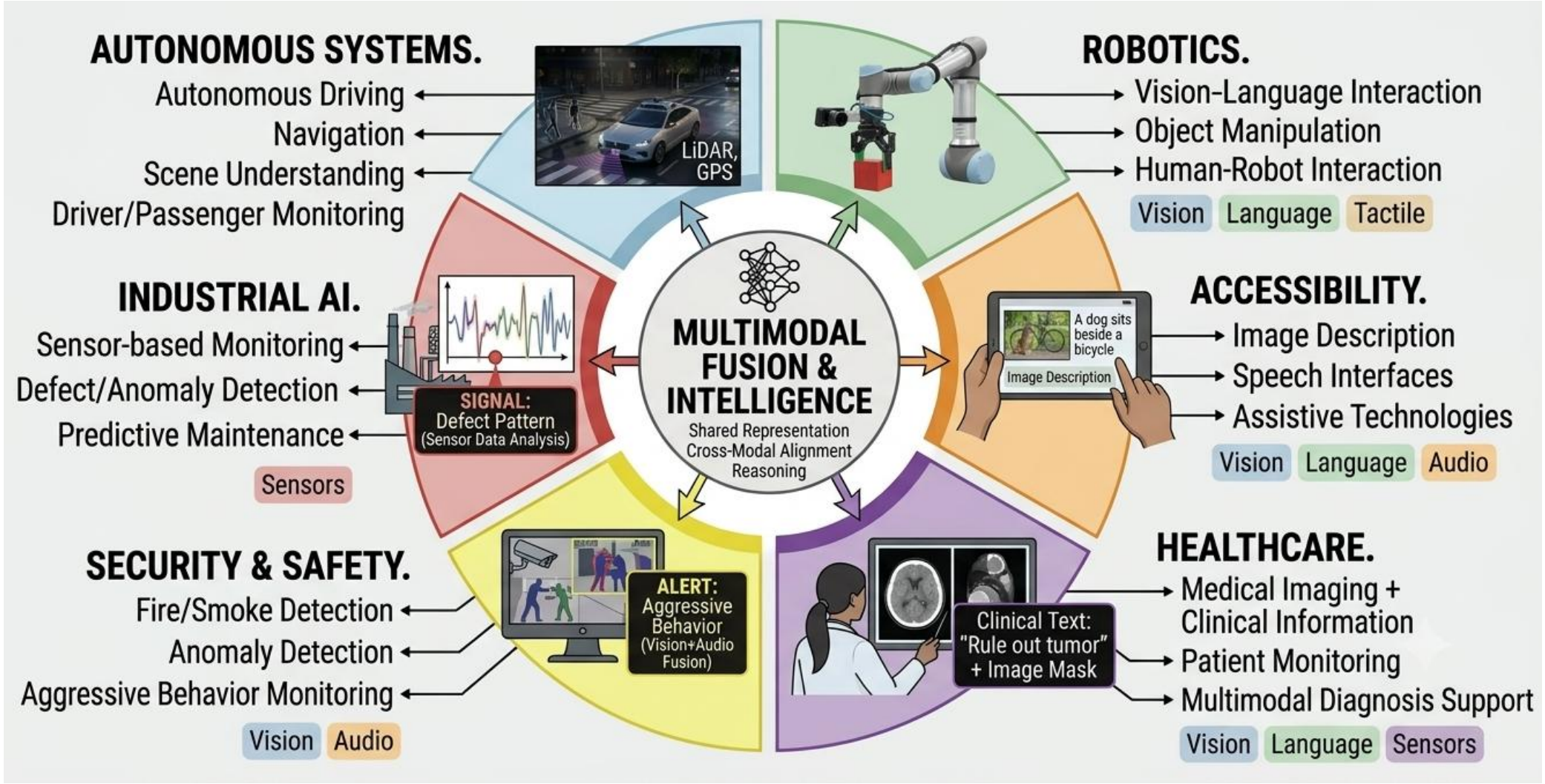
6 (iv): Understanding "Modality“ (Example)



6 (v): Application Landscape

Where Is Multimodal AI Used?

Multimodal AI spans multiple real-world domains.



6 (vi): Course Case Studies

From Theory to Real Systems

The course is structured to bridge foundational multimodal AI theory with real-world intelligent systems deployed in perception, monitoring, and decision-making tasks.

Each case study represents a progression in system complexity — from low-level perception to high-level reasoning and action.

❑ **LOW-LEVEL PERCEPTION SYSTEMS:**

1. **Image Enhancement & Restoration**

Focus:

- Improving visual quality and signal fidelity
- Noise removal, deblurring, and reconstruction
- Foundational step for all vision-based AI systems

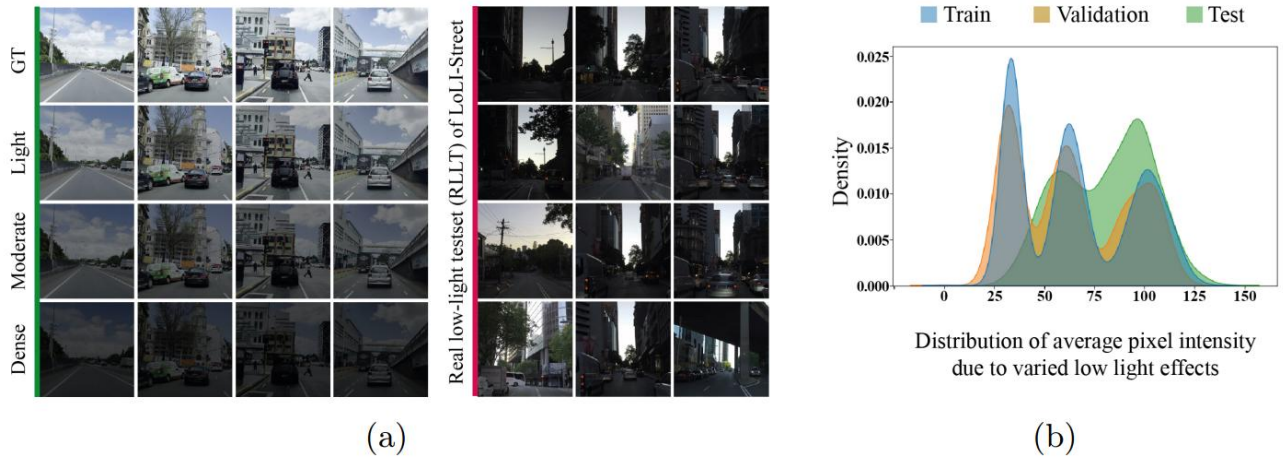


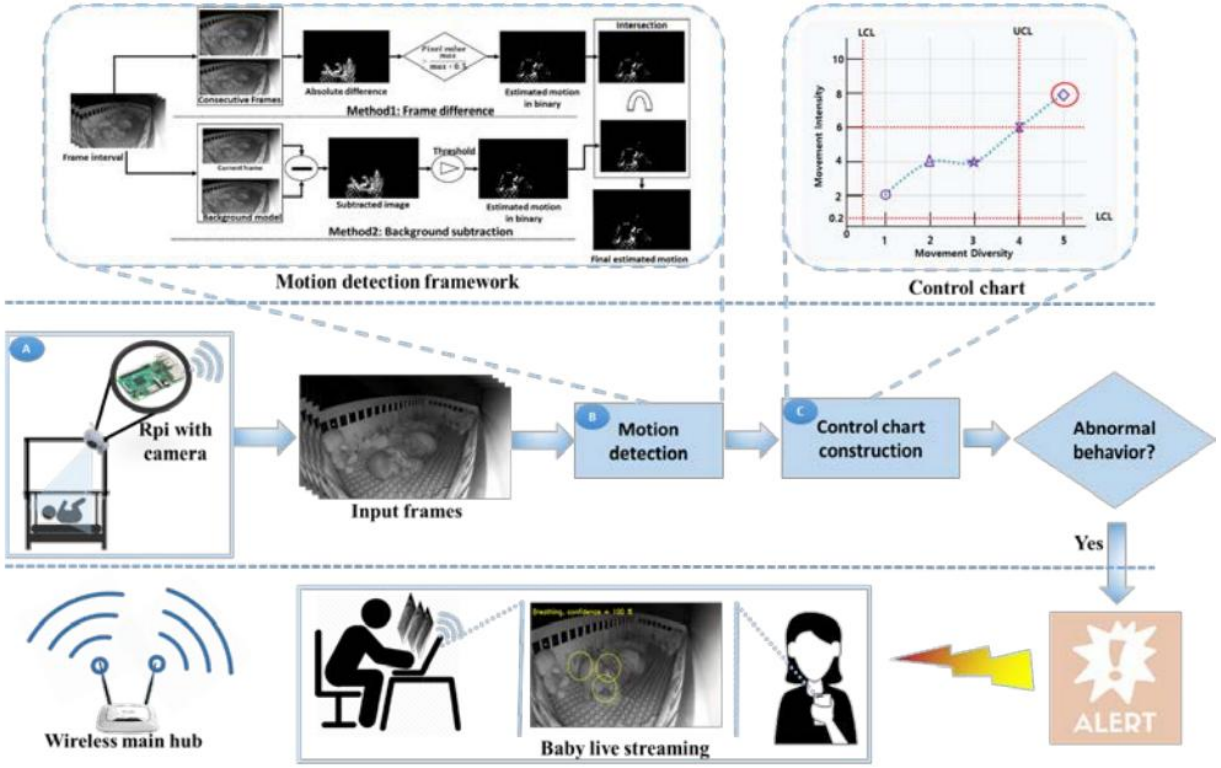
Fig. 2: (a) Sample images of LoLI-Street. **Green:** train and validation sets, **Red:** real low-light test set. (b) Distribution of the low-light images of our LoLI-Street.

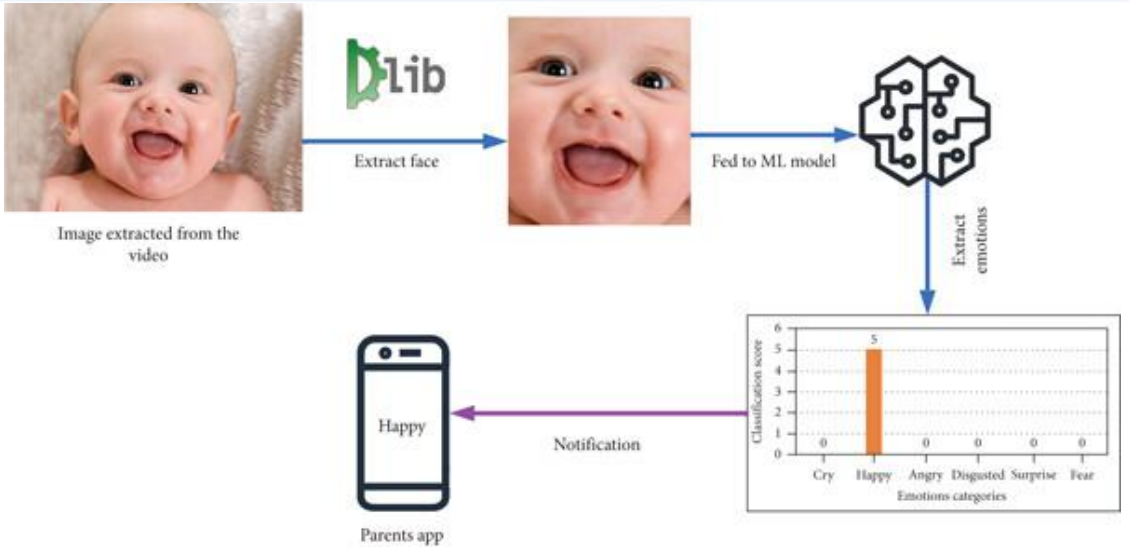
LoLI-Street: Benchmarking Low-Light Image Enhancement and Beyond, The 17th Asian Conference on Computer Vision (ACCV), 2024, Hanoi, Vietnam. BK IF=1.

Course Case Studies: From Theory to Real Systems

- **ENVIRONMENTAL UNDERSTANDING SYSTEMS**
  1. **Intelligent Baby Monitoring**

  Focus:
  - Continuous multimodal scene understanding
  - Detection of activity patterns and anomalies
  - Early-stage event recognition in dynamic environments





Proposed baby behaviour monitoring framework. (A): Input is acquired from smart camera attached to RPi, (B): motion computation where two different algorithms are executed to compute motion of baby, and (C): control chart construction from the detected motion, abnormal breathing behaviour detection based on one interval of control chart, and alert transmission in case of any abnormal breathing or motion behaviour.

Hussain, T., **Muhammad, K.**, Khan, S., Ullah, A., Lee, M. Y., & Baik, S. W. (2019). Intelligent baby behavior monitoring using embedded vision in IoT for smart healthcare centers. *Journal of Artificial Intelligence and Systems*, *1*(1), 110-124.

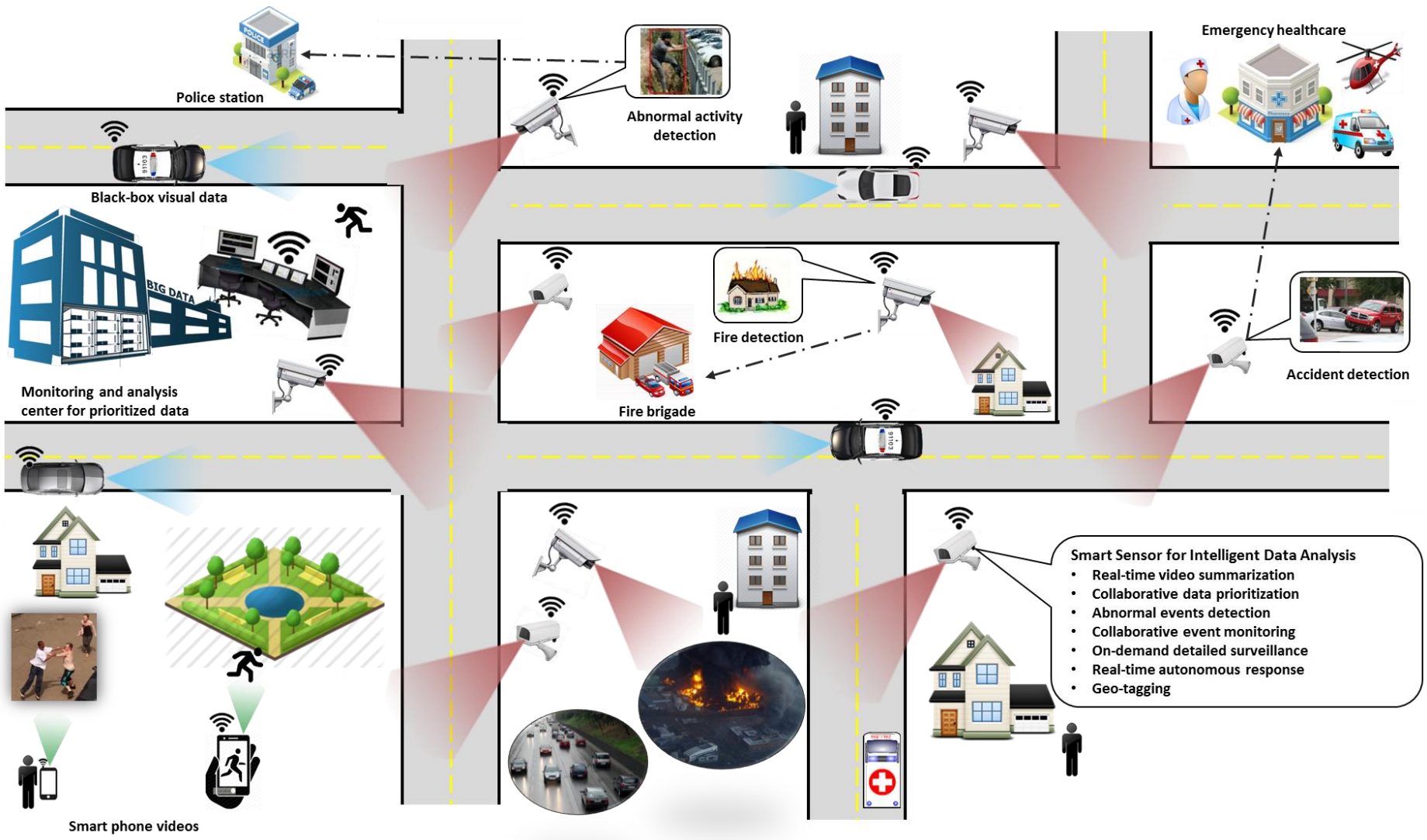
Course Case Studies: From Theory to Real Systems

❑ **ENVIRONMENTAL UNDERSTANDING SYSTEMS**

1. **Intelligent Baby Monitoring**
2. **Fire & Smoke Detection**

Focus:

- Real-time scene interpretation
- Safety-critical visual recognition
- Early warning through visual cues and environmental signals



Course Case Studies: From Theory to Real Systems

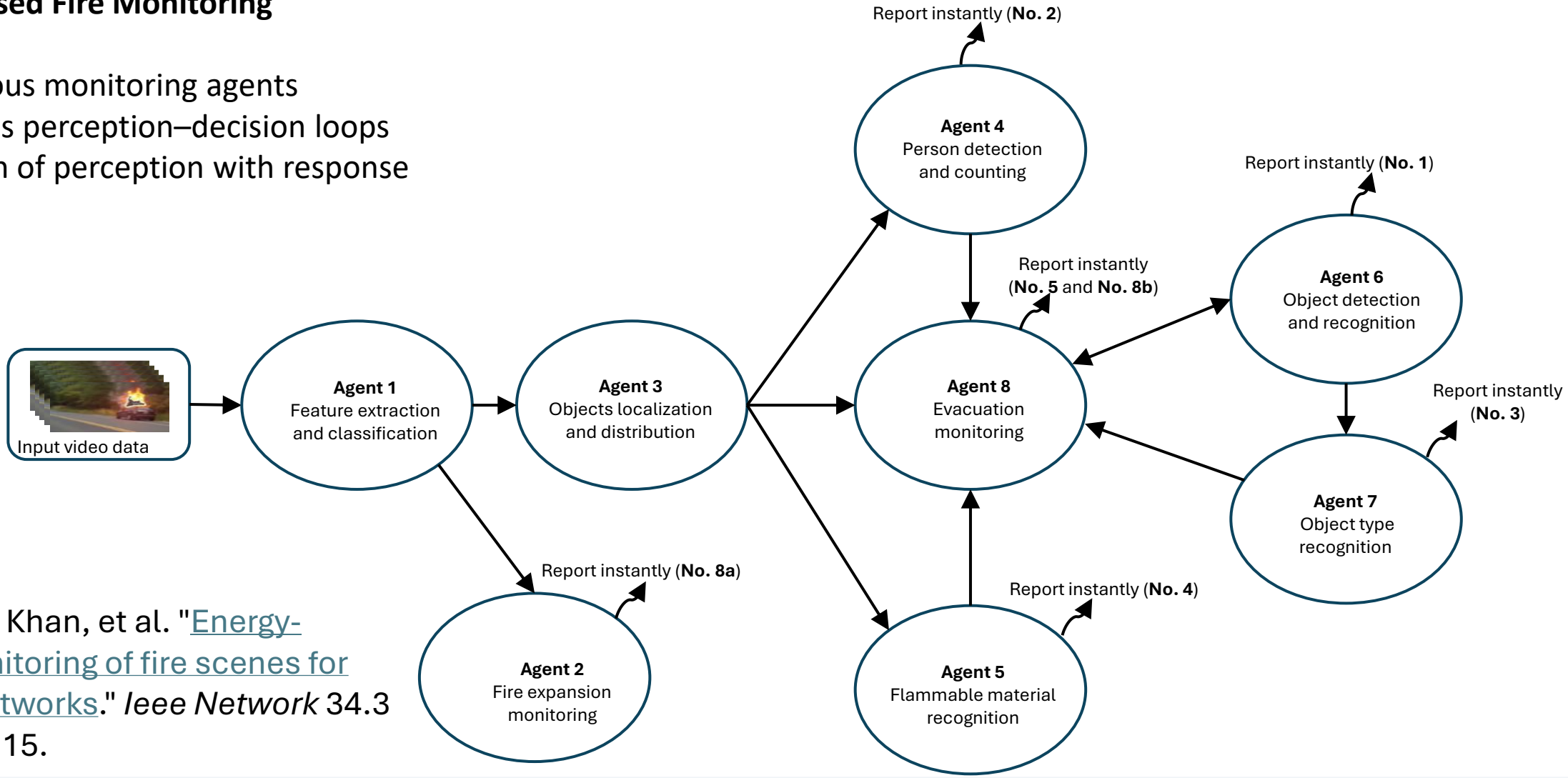
❑ **AGENTIC & DECISION-ORIENTED SYSTEMS**

1. Agent-Based Fire Monitoring

Focus:

- Autonomous monitoring agents
- Continuous perception–decision loops
- Integration of perception with response strategies

Collaborative monitoring of fire scene through agents

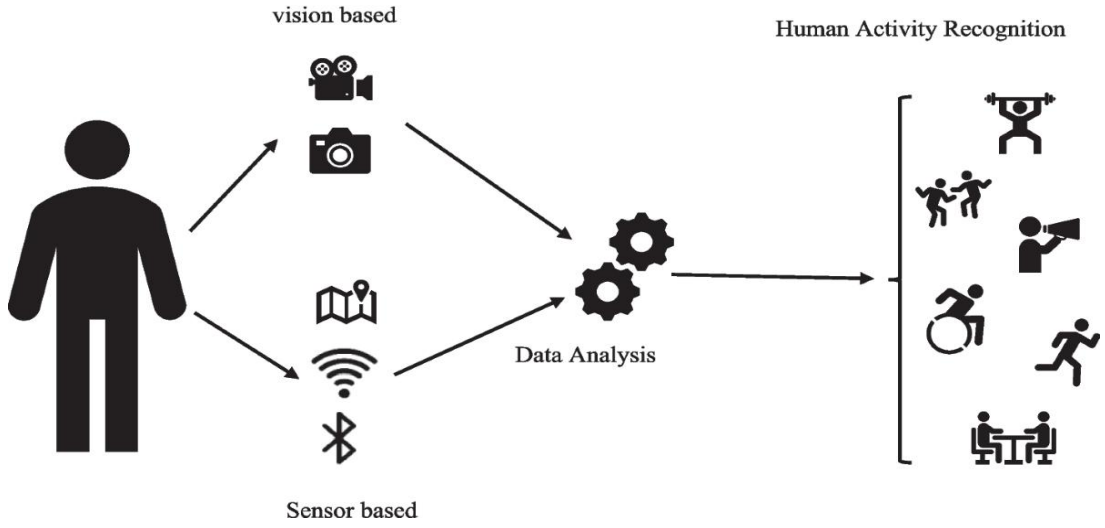


Muhammad, Khan, et al. "Energy-efficient monitoring of fire scenes for intelligent networks." *Ieee Network* 34.3 (2020): 108-115.

Course Case Studies: From Theory to Real Systems

- **BEHAVIOR & HUMAN ACTIVITY INTELLIGENCE**
  1. **Human Action Recognition**

  Focus:
  - Understanding structured human motion patterns
  - Temporal modeling of behavior
  - Multimodal sensor-based interpretation



CHILD: Child **H**azardous **I**ndoor **L**abeled **D**ataset for Activity Recognition

ACM Multimedia, 2025, BK IF=4.
https://dl.acm.org/doi/abs/10.1145/3746027.3754795



Figure 3: Example frames illustrating various activities from the CHILD dataset, categorized by risk level. Activities begin with normal activity, transition through low-risk and medium-risk, and conclude with high-risk .

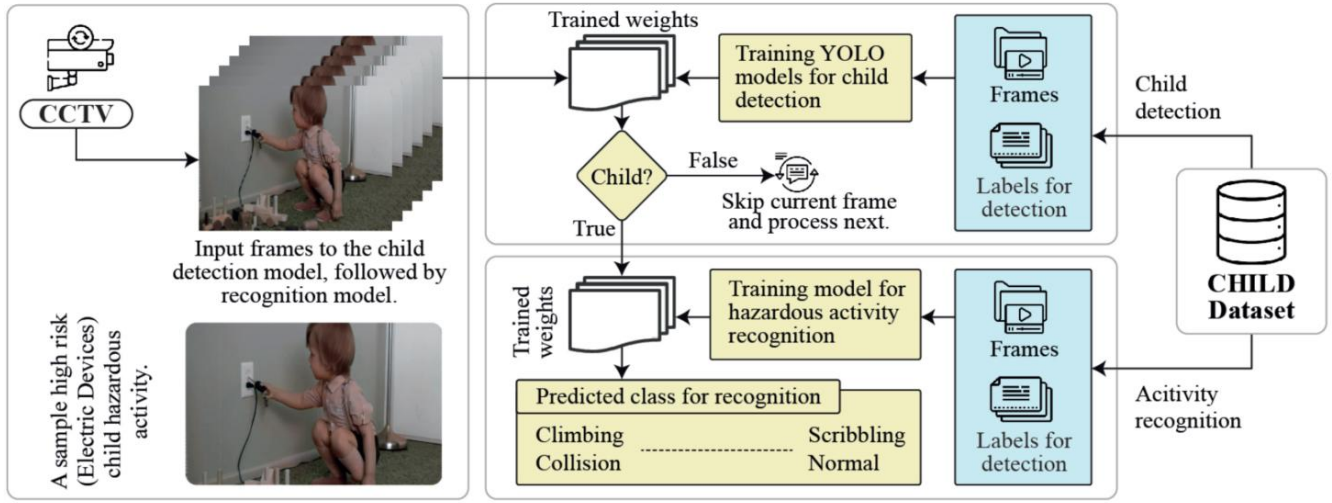


Figure 4: Proposed HazardNet: A custom framework for monitoring child-hazardous activities in indoor settings, incorporating dedicated detection and recognition models for effective performance, utilizing the newly introduced CHILD dataset.

Course Case Studies: From Theory to Real Systems

❑ **BEHAVIOR & HUMAN ACTIVITY INTELLIGENCE**

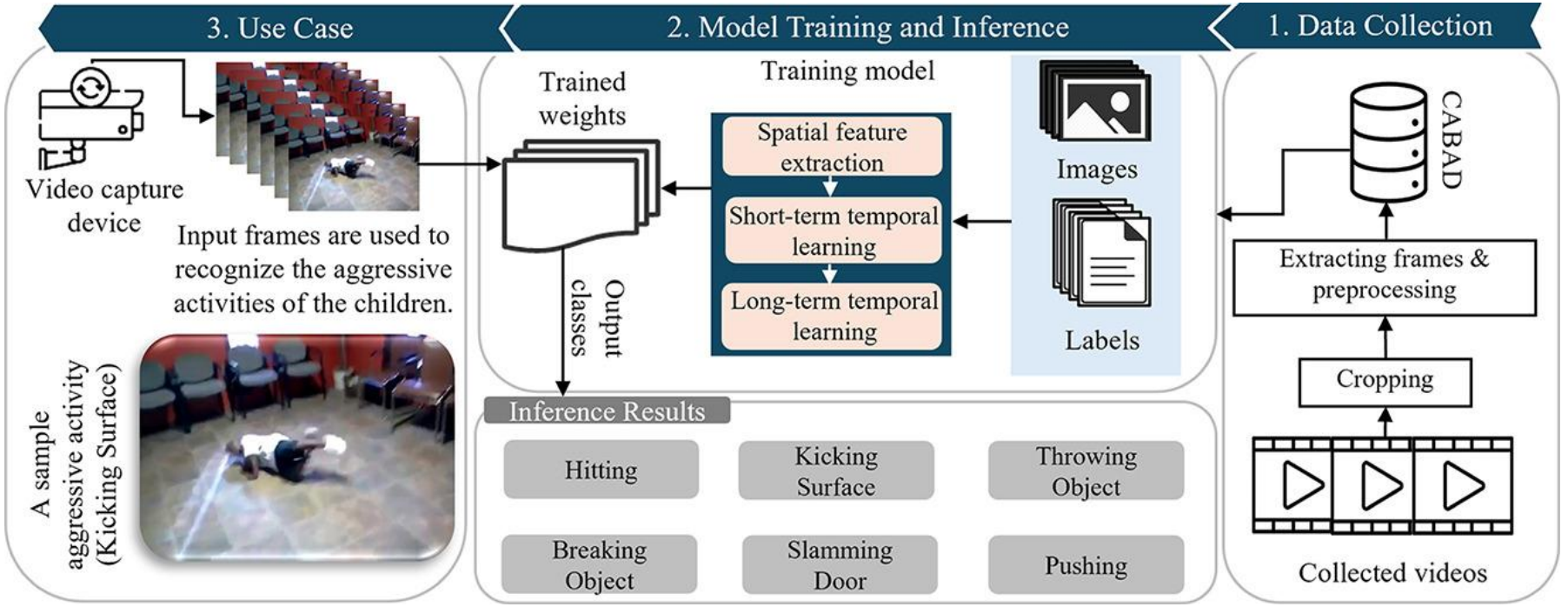
1. **Human Action Recognition**
2. **Aggressive Behavior Monitoring**

Focus:

- Complex behavior classification in real environments
- Context-aware interpretation of human actions
- Safety and surveillance applications

CABAD: A Video Dataset for Benchmarking Child Aggression Recognition

Alexandria Engineering Journal (Impact Factor: 6.2, Q1 Rank, Top 5%)



Child Aggression Behavior Analysis Dataset (CABAD) and CABA_Net

Course Case Studies: From Theory to Real Systems

ROAD-6: A Diverse Dataset for Unexpected Hazard Recognition in Autonomous Vehicles



Figure 1: Sample ROAD-6 images depicting risk levels: High (red), Moderate (orange), and Low (blue).

The 15th ACM International Conference on Multimedia Retrieval (ICMR), USA. BK IF=1.

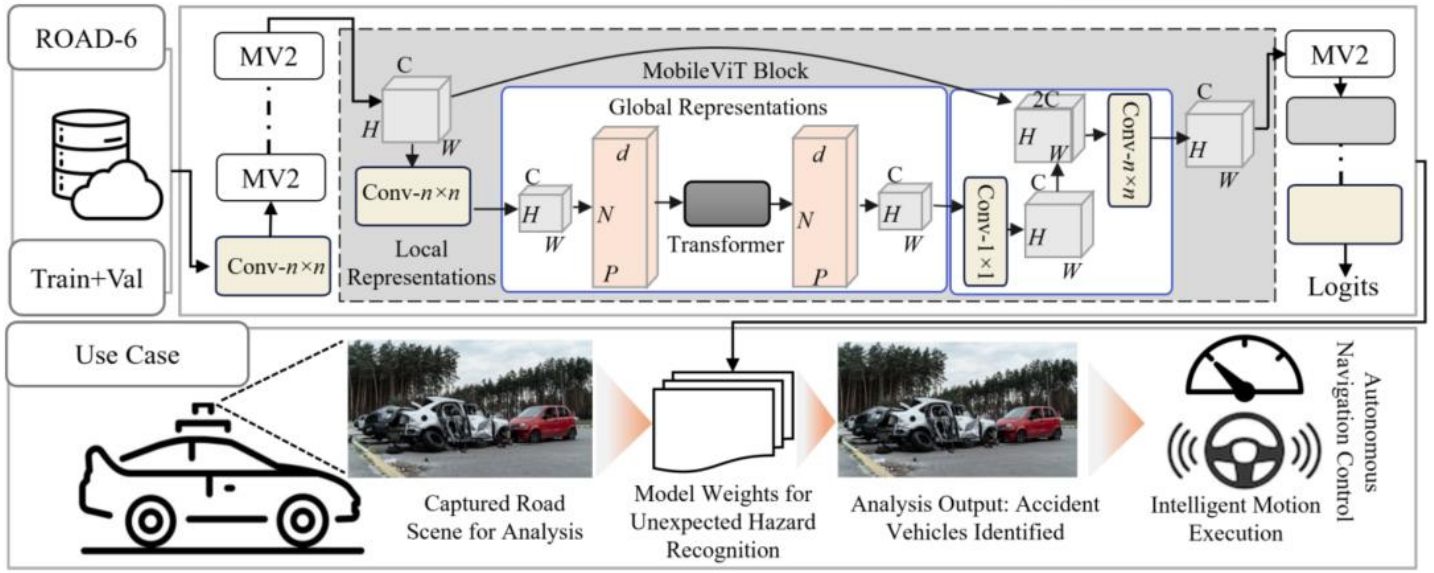


Figure 2: Use case of unexpected hazard recognition framework for AV utilizing MobileViT and the ROAD-6 dataset.

6 (vii): Course Learning Outcomes

By the end of the course, students should be able to understand these

1. Explain

Explain the fundamental concepts behind multimodal AI.

2. Represent

Understand how different modalities are represented computationally.

3. Align

Explain how information from different modalities can be associated and aligned.

4. Fuse

Understand major strategies for combining multimodal information.

5. Analyze architectures

Understand the role of neural networks, embeddings, attention, and Transformers in multimodal systems.

6. Interpret applications

Analyze how multimodal AI is used in vision, audio, video, sensors, robotics, healthcare, autonomous systems, and safety applications.

7. Reading Literature

Understand and critically discuss multimodal AI research papers and case studies.

8. Identify limitations

Recognize challenges such as noise, missing modalities, alignment errors, computational cost, bias, and reliability.

6 (viii): Course Roadmap

Phase: 1 Foundations

- Week 1: Introduction to Multimodal AI
- Week 2: Deep Learning Foundations
- Week 3: Attention & Transformers as the Multimodal Backbone
- Week 4: Vision: CNNs → Vision Transformers

Phase 2 Vision & Real-World Perception

- Week 5: Low-Level Vision & Image Enhancement
- Week 6: Intelligent Baby Monitoring
- Week 7: Fire/Smoke Detection & Scene Analysis
- Week 8: Mid Term Exam

6 (viii): Course Roadmap

Phase 3 Intelligent Multimodal Systems

- Week 9: Agent-Based Intelligent Fire Monitoring
- Week 10: Vision-Language Alignment & Open Vocabulary Object Detection
- Week 11: Audio & Speech
- Week 12: Video Understanding

Phase 4 Beyond Vision

- Week 13: Industrial Anomaly Detection
- Week 14: Sensor-Based Human Action Recognition
- Week 15: Aggressive Behavior Monitoring
- Week 16: Final Term

6 (viii): Course Roadmap

1. Professor's Topics of Research
2. Keynote Talk
   - Keynote Speaker, The 25th International Conference on Computational Science and Its Applications, Turkey, 2025
3. Intelligent Fire Scene Analysis
   - Exposure of Area
   - Thesis Tips
4. Students' Presentation
   - Topic and Contributions
   - Plan and Suggestions
5. Invited Speakers
   - Success Story and Experiences
   - Projects
   - Opportunities

7: Goals and Expectations

1. My goals

- Maximize the learning effectiveness of your time
- Provide roadmap for further exposure to related research areas
- Ensure good grade with hard work, flexibility, and equality

2. My expectations from you

- Attend and participate, when possible
- Follow up the lecture (if missed)
- Start assignments/homework/project etc. well before the deadline
- Stay tuned on the i-Campus
- Feedback about what's working for you and suggest improvements
  - You can write in an email
  - You can submit anonymously
  - Use official week 1, mid-term and final-term evaluations

Thank You

Any Question?

Feel free to contact on

khanmuhammad@g.skku.edu

Welcome to Join!